



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website www.jdcoem.ac.in E-mail info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

"A place to learn; A chance to grow"

2025-2026 (Odd Semester)

MSE-2 EXAMINATION

Course: B.Tech. In Computer Science and Engineering

Subject: LLM&PE

Max Marks: 20

Date: 06/10/2025

Year/Semester: 5th Semester (3rd Year)

Subject Code: CS5M003

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

CO1: Students will be able to Understand and apply basic and advanced prompt engineering techniques.

CO2: Students will be able to Integrate and utilize APIs to build AI applications.

CO3: Students will be able to Design and implement AI-powered chatbots and utilities.

CO4: Students will be able to Optimize generative AI applications for latency and cost.

CO5: Students will be able to Evaluate prompt performance and select appropriate fallback strategies.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	What is the purpose of authentication when using the OpenAI API?	L1/CO3	1
Q.2	Why is JSON output formatting useful when working with API responses?	L2/CO3	1
Q.3	Explain how context retention improves the performance of chatbots.	L2/CO4	1
Q.4	Explain why latency measurement is important in AI application deployment.	L2/CO5	1

SECTION B

1. All Questions are compulsory.
2. Answer in one two three sentence.

Q.5	Explain the difference between the completion and chat endpoints in the OpenAI API.	L2/CO3	2
Q.6	Suppose you are building a chatbot using LangChain. How would you use <i>chains</i> and <i>tools</i> to integrate external APIs for enhanced functionality?	L3/CO3	2
Q.7	Suppose you are developing a personal AI assistant. How would you implement stateful memory to ensure the assistant remembers user preferences across sessions?	L3/CO4	2
Q.8	Suppose your primary model fails to respond in time. How would you implement a fallback strategy using a secondary model?	L3/CO5	2



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SECTION C

1. Solve any two questions.
- | | | | |
|------|--|--------|---|
| Q.9 | Compare FAISS and ChromaDB in terms of scalability, search efficiency, and ease of integration with LLM-based applications. Which would you recommend for a large-scale semantic search system, and why? | L4/CO3 | 4 |
| Q.10 | Critically analyze how context retention and UI integration together influence the user experience in an AI-powered translator application. | L4/CO4 | 4 |
| Q.11 | In a multi-model orchestration setup, analyse the trade-offs between using a cheaper but less accurate model versus a more costly but highly accurate one. | L4/CO5 | 4 |